

▷ LIFE ON AIR

Epiphytes, or air plants, like this *Cattleya* orchid, require no soil. Instead, they anchor themselves to trees, drawing water and nutrients from moist air.



▷ CLOSED CANOPY

Most rain forest plants and animals live in the canopy, where trees compete for sunlight via maximum leaf growth and spread. The overlapping branches form an almost impenetrable "roof."

**Brazil nut tree**

This emergent tree reaches heights of 130-165 ft (40-50 m). Most Brazil nuts come from wild rain forest trees.

Rubber tree

A rubber tree grows very fast so will soon establish itself if a gap appears in the canopy.

Cacao tree

The cacao tree grows in the understory in the shade of other trees. Each of its pods contains up to 60 cocoa beans.

Leaf litter

Microorganisms feed on and recycle decaying plant material lying on the forest floor.

Shallow roots

An extensive shallow root system allows a tree to absorb vital topsoil nutrients before they are washed away by rain.

Topsoil

Rain forests have just a thin layer of topsoil. The soil contains all available organic matter and nutrients for plant growth.

Buttress roots

These large roots grow at angles from the trunk to stabilize a tall tree in poor-quality soil.

▷ FOOD ON THE FLOOR

Some rain forest trees, like the Brazil nut, rely on ground-dwelling animals to spread their seeds. The agouti gnaws holes in tough, fallen seed pods to reach the nuts inside. It buries some but often forgets them, so new Brazil nut trees grow instead.

